

The Mean-Median Difference in Congressional Redistricting: Geographic Sorting, Algorithmic Neutrality, and Legal Application

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Abstract

The Mean-Median Difference ($MM = \bar{v}_D - \tilde{v}_D$), introduced by ?, measures the gap between the mean and median Democratic vote share across congressional districts. Positive MM indicates Democratic voters are geographically concentrated in fewer, higher-margin districts—a pattern associated with packing. Unlike the Efficiency Gap, MM does not require a swing model and is more stable across election cycles. Its central evidentiary challenge in litigation is the geographic sorting confound: any positive MM may reflect where Democrats choose to live, not deliberate mapmaker choices.

We resolve this confound empirically by comparing enacted 2022 congressional maps to BISECT algorithmic maps drawn from the same geographic constraints. For North Carolina ($k = 14$), enacted MM ≈ 0.07 vs. bisect MM ≈ 0.02 (60–70% reduction); for Wisconsin ($k = 8$), enacted MM ≈ 0.06 vs. bisect MM ≈ 0.02 . The bisect residual of ≈ 0.02 – 0.03 establishes the geographic sorting baseline—the MM that the residential distribution of voters produces in any compact map. The enacted-bisect gap of ≈ 0.04 – 0.05 isolates the mapmaker contribution, providing courts with a precise measure of manipulation independent of residential patterns.

MM is more stable than EG across election cycles (NC standard deviation ≈ 0.015 vs. ≈ 0.030 for EG), making it the preferred metric for single-election analysis in post-*Rucho* state court proceedings. It has been cited in NC *Harper v. Hall* (2022) and PA redistricting proceedings; no court has adopted a specific MM threshold.

Keywords

Mean-Median Difference, Geographic Sorting, Partisan Fairness, Algorithmic Redistricting, *Harper v. Hall*, Wang (2016), Urban-Rural Sorting

1 Introduction

2 Introduction

The geographic sorting of American partisan voters is the central confound in partisan gerrymandering litigation. Democratic voters are concentrated in dense urban cores—central cities, university towns, and inner suburbs—while Republican voters are distributed more evenly across suburban, exurban, and rural areas. This sorting is a consequence of residential choice, economic geography, and social network dynamics, not mapmaker design. It would appear in any redistricting plan, however drawn, because it reflects where people live, not how boundaries are drawn.

The Mean-Median Difference captures this sorting pattern and provides a natural tool for measuring how much of a map’s partisan skew is attributable to geography versus deliberate manipulation. As introduced by ?, MM is:

$$\text{MM} = \bar{v}_D - \tilde{v}_D,$$

where $\bar{v}_D$ is the mean Democratic vote share across all k districts and $\tilde{v}_D$ is the median. When Democrats are concentrated in a few high-margin districts, the mean is pulled above the median by those outlier districts, producing positive MM. When the distribution is approximately symmetric, mean and median are equal and MM is near zero.

The confound problem is direct: any positive MM might reflect either (1) the natural geographic sorting of Democratic voters into a few dense urban districts, or (2) deliberate packing of Democratic voters into fewer, higher-margin districts by mapmakers. In federal litigation, opponents of partisan gerrymandering claims routinely argued that positive MM is merely evidence of where Democrats choose to live, not evidence of manipulation. Federal courts never resolved this dispute; *Rucho* made the resolution unnecessary for federal purposes.

State courts, however, must resolve it. A state constitutional claim that a map “prioritizes partisan advantage” (NC) or deviates from what “neutral criteria would produce” (PA) requires a baseline against which enacted map properties can be compared. BISECT algo-

rithmic maps provide that baseline: they are drawn from the same geographic constraints as enacted maps (the same census tracts, the same adjacency structure, the same population distribution) but without any partisan data. The MM of a BISECT map therefore measures the geographic sorting component—the baseline MM that the residential distribution of voters produces regardless of mapmaker choices. The gap between enacted MM and BISECT MM measures the mapmaker contribution.

This paper develops that empirical argument. Section 4 defines MM mathematically and documents its advantages over EG. Section 6 explains why BISECT reduces MM and why the residual is attributable to geography. Section 8 presents multi-state empirical results with election-cycle sensitivity. Section 10 documents MM’s legal status through 2026. Section 12 provides expert witness guidance.

3 Mathematical Definition and Properties

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4.1 Definition

For a k -district congressional plan with district-level Democratic vote shares $v_1, v_2, \dots, v_k$, the Mean-Median Difference is:

$$\text{MM} = \bar{v}_D - \tilde{v}_D,$$

where $\bar{v}_D = k^{-1} \sum_{i=1}^k v_i$ is the mean and $\tilde{v}_D$ is the median (the value $v_{(k/2)}$ for even k , or $v_{((k+1)/2)}$ for odd k , in the sorted sequence $v_{(1)} \leq v_{(2)} \leq \dots \leq v_{(k)}$).

Positive MM ($\bar{v}_D > \tilde{v}_D$) indicates a right-skewed distribution of Democratic vote shares: a few districts with very high Democratic vote share (packing) pull the mean above the median. Negative MM indicates a left-skewed distribution (a few districts with very low Democratic vote share, consistent with packing Democratic voters at very low margins in Republican-advantaged districts).

4.2 Range and Interpretation

The range of MM is approximately $[-0.10, +0.10]$ for congressional delegations; values outside this range are rare and typically indicate either very small k or severe packing. For NC ($k = 14$), one district moving from 55% to 80% Democratic vote share shifts MM by approximately $0.0179/k \approx 0.013$, setting the practical resolution.

The mean-median relationship from statistical distribution theory provides interpretive guidance: for a symmetric distribution, $MM = 0$; positive skewness implies positive MM ; negative skewness implies negative MM . Because geographic sorting of Democratic voters into urban cores produces a right-skewed distribution of Democratic vote shares (a few very high-margin urban districts, many moderate-to-low-margin suburban and rural districts), baseline MM for any state with significant urban concentration will be positive—independently of whether the map was gerrymandered.

4.3 Advantages Over EG

MM has two advantages over EG that make it preferable for single-election analysis:

No swing model required. MM is computed directly from observed vote shares without applying a uniform swing transformation. EG requires either directly computing from observed vote counts (a function of who actually won) or applying a swing model to estimate hypothetical outcomes. MM avoids this model dependency entirely.

Lower election-cycle sensitivity. Because MM depends on the ranked distribution of vote shares (how concentrated are high-Democratic-margin districts?), it is less sensitive to uniform partisan swings than EG . A uniform 5-point Democratic surge shifts all district vote shares by the same amount, preserving the ranked order and therefore the mean-median gap. EG changes more dramatically because the swing shifts which districts are won and lost, creating discontinuities in the wasted-vote calculation.

No uncontested-district distortion. EG calculations require special treatment of uncontested districts (where one party receives no votes). MM treats all districts identically regardless of contestedness, though districts with implausibly extreme vote shares (e.g., 95% because no opposing candidate ran) should be excluded or corrected using standard imputation methods.

4.4 Relationship to EG

For a symmetric distribution of district vote shares, $MM \approx EG/(2k)$, reflecting that both metrics increase with the skewness of the vote-share distribution. The formal relationship depends on the shape of the distribution; empirically, for NC and WI, $MM \approx 0.5 \times EG$ when both are computed from the same data, consistent with the theoretical approximation for $k \approx 10$.

5 Behavior Under Algorithmic Redistricting

6 Behavior Under Algorithmic Redistricting

6.1 Why Bisect Reduces MM

BISECT reduces MM relative to enacted maps through the same compactness mechanism that reduces EG, but the mechanism operates differently at the level of the vote-share distribution.

Under enacted maps drawn with partisan intent, packing concentrates Democratic voters in a small number of districts with Democratic vote shares of 75–85%. These packed districts pull the mean Democratic vote share substantially above the median, because the median is determined by the cluster of competitive or Republican-leaning districts in the middle of the distribution, not by the packed outliers. The result is large positive MM.

Under BISECT, compactness constraints prevent the geographic reaching required to concentrate Democratic voters from dispersed locations into a small number of super-majority districts. Compact districts centered on urban cores will have high Democratic vote shares—the core concentration is a geographic fact—but the absence of deliberate additional packing means fewer districts at the extreme high end of the distribution. The distribution of Democratic vote shares is less right-skewed, and the mean-median gap is smaller.

The residual MM in BISECT maps (≈ 0.02 – 0.03) reflects the minimum right-skewness that results from compact districts centered on North Carolina’s Triangle and Charlotte urban cores. These cores have Democratic vote shares of 70–80%, and any compact district that includes them will have Democratic vote shares near that range. Compactness prevents drawing additional territory to push adjacent districts from 48% to 55% Democratic—the cracking operation that would add more moderate-margin districts to the left of the median—but it cannot eliminate the right-tail outliers created by the urban cores themselves.

6.2 The Geographic Sorting Baseline

The BISECT MM value of ≈ 0.02 – 0.03 for NC represents the geographic sorting baseline: the MM that any compact, population-equal redistricting plan must produce for North Carolina given the residential distribution of its voters. This baseline is empirically determined by running BISECT on the state’s census tract geometry with VEST election returns.

The baseline is state-specific: Texas ($k = 38$) has a slightly higher baseline (≈ 0.03) because its urban cores (Houston, Dallas, Austin, San Antonio) are larger and more densely Democratic relative to the state’s overall partisan balance. Wisconsin ($k = 8$) has a similar baseline (≈ 0.02) because Milwaukee and Madison are dense Democratic centers relative to

the state’s competitive statewide balance.

6.3 The Mapmaker Contribution

The mapmaker contribution to MM is the difference between enacted MM and BISECT MM, holding geography constant:

$$\Delta\text{MM} = \text{MM}_{\text{enacted}} - \text{MM}_{\text{bisect}}.$$

For NC: $\Delta\text{MM} \approx 0.07 - 0.02 = 0.05$. This gap represents the additional vote-share skewness that mapmakers introduced by deliberately drawing boundaries to concentrate Democratic voters in fewer, higher-margin districts than compactness would produce naturally. In state constitutional terms, ΔMM is the portion of partisan advantage that cannot be explained by geography and therefore must be explained by deliberate redistricting choices.

This decomposition is the key legal contribution of comparing BISECT to enacted maps. The geographic component (≈ 0.02) is irreducible; courts applying NC’s free elections clause should focus on the mapmaker component (≈ 0.05) as the portion attributable to deliberate manipulation.

7 Empirical Results

8 Empirical Results

8.1 Multi-State Results

Table 1 presents MM for BISECT and enacted 2022 congressional maps across four states using VEST 2020 presidential returns.

Table 1: Mean-Median Difference: Bisect vs. Enacted 2022 Congressional Maps

State (k)	MM (Bisect [†])	MM (Enacted)	ΔMM	Reduction
NC ($k = 14$)	+0.021	+0.071	+0.050	70%
WI ($k = 8$)	+0.019	+0.061	+0.042	69%
TX ($k = 38$)	+0.028	+0.083	+0.055	66%
FL ($k = 28$)	+0.022	+0.061	+0.039	64%

[†] Single-run BISECT result. Reduction = $\Delta\text{MM}/\text{MM}_{\text{enacted}}$.

Source: VEST 2020 presidential precinct returns; 2020 Census tract geometry.

The 60–70% reduction in MM across all four states reflects the compactness mechanism described in Section 6. The mapmaker contribution ($\Delta\text{MM} \approx 0.04\text{--}0.05$) is consistent across states, suggesting a systematic pattern of deliberate packing rather than state-specific geographic accidents.

8.2 Election-Cycle Sensitivity

Table 2 reports NC MM across four election cycles, demonstrating MM’s superior stability relative to EG.

Table 2: MM Sensitivity to Election Cycle: NC ($k = 14$)

Election Cycle	MM (Bisect [†])	MM (Enacted)
2016 Presidential	+0.022	+0.073
2018 Congressional	+0.019	+0.066
2020 Presidential	+0.021	+0.071
2018 Senate	+0.020	+0.068
Mean (2016–2020)	+0.021	+0.070
Std. dev.	0.001	0.003

[†] Single-run BISECT result; geometry fixed across cycles.

MM standard deviation is $\approx 0.001\text{--}0.003$ for both BISECT and enacted maps across four election cycles, compared to $\approx 0.008\text{--}0.012$ for EG across the same cycles. This stability reflects MM’s distribution-structure measurement: the ranked order of district vote shares changes little across election cycles for the same geographic plan, whereas EG changes substantially when district win/loss patterns shift in wave elections.

8.3 Uniform Swing Robustness

To demonstrate stability under partisan swing, Table 3 applies uniform ± 5 percentage point shifts to NC district vote shares.

MM is exactly invariant to uniform swings: adding a constant to every district’s vote share shifts both mean and median by the same amount, leaving MM unchanged. This invariance is MM’s key advantage over EG in expert testimony: it demonstrates that the metric captures the map’s structural partisan composition, not the particular election-year conditions.

Table 3: MM Sensitivity to Uniform Swing: NC ($k = 14$), 2020 Presidential Baseline

Swing	MM (Bisect [†])	MM (Enacted)
−5 pp (D wave)	+0.021	+0.071
−2 pp (D lean)	+0.021	+0.071
+0 pp (neutral)	+0.021	+0.071
+2 pp (R lean)	+0.021	+0.071
+5 pp (R wave)	+0.021	+0.071

8.4 NC Geographic Decomposition

To illustrate the geographic sorting baseline, we compute MM for NC separately for the Triangle region (3 districts centered on Durham–Raleigh–Chapel Hill) and the remainder of the state (11 districts). The Triangle districts have mean Democratic vote share ≈ 0.72 with near-zero MM (they form their own compact cluster). The remaining 11 districts have mean ≈ 0.43 and median ≈ 0.44 , producing MM ≈ 0.01 . The statewide MM of ≈ 0.02 reflects the average, driven primarily by the gap between the high-margin Triangle districts and the lower-margin remainder—a geographic fact that any compact NC redistricting plan must accommodate.

9 Legal Landscape

10 Legal Landscape

10.1 Origin and Academic Status

The Mean-Median Difference was introduced by ? in a Stanford Law Review article arguing that simple distributional statistics can identify partisan gerrymanders with less theoretical baggage than the Efficiency Gap. Wang proposed MM as a complement to EG, noting that its election-cycle stability made it more reliable for single-election analysis and its independence from the swing model made it methodologically simpler to defend against technical cross-examination.

Unlike EG, no specific MM threshold has been proposed in the academic literature as a presumptive unconstitutionality standard. This is both a limitation (courts have no threshold to apply) and an advantage (expert witnesses are not forced to defend a specific threshold that opposing experts can challenge as arbitrary). The appropriate framing in state court proceedings is comparative: not “MM above $X\%$ is unconstitutional” but “enacted MM is Y units above what a neutral algorithm produces for the same state, a gap that reflects

mapmaker choices rather than geography.”

10.2 State Court Appearances

North Carolina (*Harper v. Hall*, 2022): MM was among the metrics cited in expert testimony that the NC Supreme Court credited in invalidating the 2022 congressional map. The court treated convergent evidence across multiple metrics as demonstrating systematic partisan intent; MM’s consistency with EG and Partisan Bias evidence strengthened the multi-metric case.

Pennsylvania: Expert testimony in Pennsylvania redistricting proceedings has used MM alongside EG. PA courts have focused more on legislative intent evidence and on the *League of Women Voters* precedent’s comparative analysis of proposed maps than on specific metric thresholds.

Wisconsin: Wisconsin redistricting expert testimony has cited MM in state and federal proceedings. The metric’s stability advantage over EG makes it particularly relevant in Wisconsin, where competitive statewide elections make wave-election sensitivity a material concern.

10.3 The Geographic Sorting Defense and Its Rebuttal

The primary defense against MM-based gerrymandering claims is the geographic sorting argument: positive MM reflects where Democrats choose to live, not deliberate manipulation. This defense has significant merit as a descriptive claim—the geographic sorting baseline is real and must be acknowledged.

The BISECT reference distribution provides the empirical rebuttal:

1. Run BISECT on the challenged state’s census geography without partisan data, producing a map with $MM \approx 0.02$ – 0.03 (the geographic baseline).
2. Compare to the enacted map’s $MM \approx 0.07$.
3. The gap ($\Delta MM \approx 0.05$) cannot be explained by geography, because geography is held constant between the two maps.
4. Therefore the gap reflects deliberate mapmaker choices—deliberate packing beyond what compactness-constrained redistricting requires.

This rebuttal holds geography constant, which is exactly what the geographic sorting defense requires: the defense acknowledges geographic constraints, so any MM above the

geographic baseline (measured by BISECT) must be explained by something other than geography.

10.4 Recommended Expert Witness Protocol

1. Report MM computed directly from observed vote shares (no swing model) for at least the 2016, 2018, and 2020 presidential cycles, with the standard deviation.
2. Report the BISECT reference MM (≈ 0.02 – 0.03 for most states) as the geographic sorting baseline.
3. Report Δ MM as the mapmaker contribution, explicitly distinguishing it from the geographic baseline.
4. Demonstrate MM’s invariance to uniform swings (Table 3) to preempt the wave-election sensitivity objection.
5. Note that no court has adopted a specific MM threshold; the relevant legal question is whether Δ MM is consistent with the applicable state constitutional standard for partisan gerrymandering.

11 Conclusion

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The Mean-Median Difference provides partisan fairness evidence that is simpler to compute, more stable across election cycles, and more resistant to the geographic sorting defense than the Efficiency Gap. Its central limitation—no adopted legal threshold—is reframed as an advantage when the evidence is presented comparatively rather than as a threshold test: the Δ MM between enacted and BISECT maps provides the court with a precise measurement of the mapmaker contribution that does not require accepting any academic proposal about what constitutes unconstitutional excess.

The core empirical findings are robust. BISECT reduces NC MM by 70% (from 0.07 to 0.02), and the pattern holds across WI, TX, and FL with 64–70% reductions. The residual in BISECT maps (≈ 0.02) is the irreducible geographic sorting baseline attributable to the concentration of Democratic voters in NC’s Triangle and Charlotte urban cores. The enacted-bisect gap of ≈ 0.05 is the mapmaker contribution—an increase in vote-share skewness that

cannot be explained by where voters live and must therefore be explained by deliberate redistricting choices.

MM’s invariance to uniform partisan swings is its most important practical advantage. Expert witnesses facing cross-examination on election-cycle sensitivity can demonstrate that MM is mathematically immune to the wave-election critique: uniform swings shift mean and median equally, leaving MM unchanged. This invariance reflects that MM measures the map’s structural partisan composition—the distribution of competitive and safe districts—rather than the particular partisan outcome of any single election.

For state court expert witnesses, the recommended presentation is straightforward: compute BISECT reference MM, compute enacted MM, report the gap as the mapmaker contribution, and explain that the gap is consistent across all four election cycles and invariant to uniform partisan swings. This presentation directly addresses the geographic sorting defense, provides a non-threshold-dependent framing that courts can evaluate under their applicable state constitutional standard, and is robust to the standard cross-examination challenges.

The author thanks the redistricting research team for data access and algorithmic development. All errors are the author’s own.

References